

Experiment 5.2: Forces and torques in equilibrium

Notes

Aim

In this investigation you will verify the conditions you need for a body to be in equilibrium. You will do this by investigating the relationship between

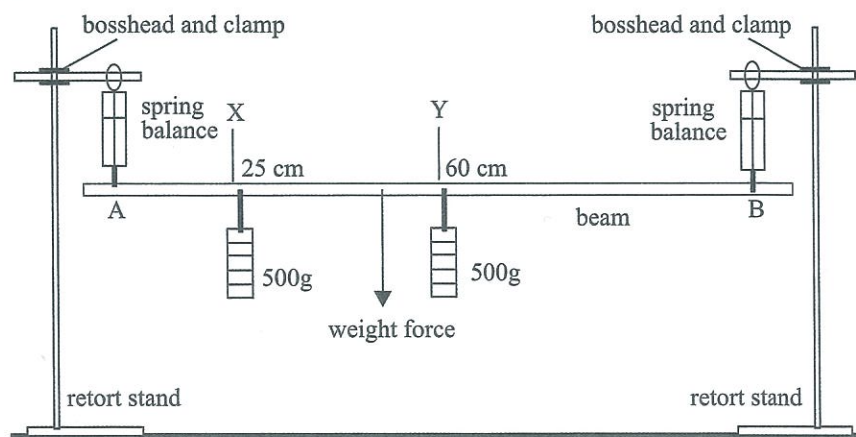
- forces that act on a body that is in equilibrium
- moments that act on a body that is in equilibrium

Apparatus

- one metre rule to act as a beam
- two sets of slotted 50 g masses
- two retort stands with boss heads and clamps
- two 0-10 N spring balances
- wire, string or paper clips for suspending masses from the metre rule and beam from spring balances

Pre-lab

- In this experiment you will use two spring balances to support a beam on which you hang two weights at different points along its length.
- Measure the weight of the beam using one of the spring balances and record your value in the table.



Apparatus for Experiment 5.2

- Set up the equipment as shown with attention to the following points:
 - a) Position the retort stands with the longer parts of their bases towards each other.
 - b) Position the spring balances on the clamps so that a vertical line through them passes within the area of the base of the stand. The spring balances should be near to and equidistant from each end of the beam.
 - c) The spring balances must hang vertically before you take a reading.
 - d) Make sure the beam balances horizontally before you take a reading.

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Lab notes

- Once the beam is in equilibrium record the reading on each spring balance and the position of the weights in the table below. Record the weights in newtons and the positions in metres measured from the spring balance at A.
- Change the positions and sizes of each weight to obtain another two sets of measurements. Use different weight values for the two weights in both trials.
- Complete the column for trial 1 in table 1. The distances should be measured from the position Y.

Note: The spring balances exert upward forces on the beam while the slotted weights and the weight of the beam exert downward forces.

Post-lab discussion

Weight of beam = _____ g				
Measurements taken	Symbol	Trial 1	Trial 2	Trial 3
Upward force at A (N)	F_A			
Upward force at B (N)	F_B			
Distance from A to B (m)	b			
Downward force at X (N)	F_X			
Distance from A to X (m)	x			
Downward weight force of beam (N)	F_w			
Distance of centre of gravity from A (m)	cg			
Downward force at Y (N)	F_Y			
Distance from A to Y (m)	y			
Distance from Y to B (m)	B			
Distance of centre of gravity from Y (m)	CG			
Distance from X to Y (m)	X			
Distance from Y to A (m)	A			

Table 1

- For *each* trial calculate $F_A + F_B$ and $F_X + F_Y + F_w$ and record your results in table 2. What do you notice about these values in each trial?
- For *each* trial calculate the sum of the clockwise moments about A and the anticlockwise moment about A and record your results in table 2. What do you notice about these values in each trial?
- For *trial 1 only*, calculate the clockwise moment about Y and the sum of the anticlockwise moments about Y and record your results in table 2. What do you notice about these values?
- Write a statement about vertical forces that summarises your findings in this experiment.
- Write a statement about moments that summarises your findings in this experiment.
- Write a statement about the position on the beam about which moments can be taken when a system is in equilibrium.
- Explain why the spring balances have to be attached equidistant from the ends of the uniform beam.
- What assumption did you make about the centre of mass of the beam in this experiment?
- Explain why it is important to position the retort stands with the longer ends of their bases towards each other.
- Explain why the spring balances must be positioned over the bases of the stands.
- Explain why the spring balances must both be vertical when you take readings from them.

Measurements taken	Symbol	Trial 1	Trial 2	Trial 3
Sum of upward force on beam (N)	$F_A + F_B$			
Sum of downward force on beam (N)	$F_X + F_Y + F_w$			
Sum of clockwise moments about A (N m)	ΣCM_A			
Anticlockwise moment about A (N m)	ΣACM_A			
Clockwise moment about Y	ΣCM_Y			
Sum of anticlockwise moments about Y (N m)	ΣACM_Y			

Table 2